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Existence and uniqueness of traveling front solutions for the partially congested Navier–Stokes equations in Eulerian coordinates

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Abstract

We study traveling front solutions for one-dimensional compressible Navier–Stokes equations with a singular pressure modeling congestion effects. Motivated by earlier results obtained in Lagrangian coordinates, we analyze the problem in the Eulerian framework. We first consider the free–congested limit system and establish existence, uniqueness (up to translation), and explicit formulas for partially congested traveling fronts. These profiles consist of a fully congested region connected to a free flow through a smooth viscous layer. We then address the singular pressure Navier–Stokes system and prove existence and uniqueness of traveling fronts under suitable Rankine–Hugoniot conditions. An ODE formulation is derived to characterize the profiles and their monotonicity properties. Finally, we show that, as the singular pressure parameter tends to zero, the Eulerian traveling fronts converge to those of the free–congested system, providing a rigorous asymptotic link between the two models.

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1 Introduction

In [3], the authors are concerned with the analysis of viscous shock waves for the following compressible Navier-Stokes system written in Lagrangian mass coordinates $(t, x) \in \mathbb{R}_+ \times \mathbb{R}$ (we refer to the appendix for details concerning the passage from Eulerian coordinates to Lagrangian mass coordinates)

$$\partial_t v - \partial_x u = 0, \quad (1.1a)$$

$$\partial_t u + \partial_x p_\varepsilon(v) - \mu \partial_x \left(\frac{1}{v} \partial_x u \right) = 0, \quad (1.1b)$$

where v is the specific volume, u is the velocity, μ is a viscosity coefficient and p_ε is the pressure. The latter is assumed to be singular close to the critical value $v^* = 1$,

$$p_\varepsilon(v) = \frac{\varepsilon}{(v-1)^\gamma} \quad \gamma > 0, \quad (1.2)$$

with $\varepsilon \ll 1$. We supplement system (1.1) with initial data

$$(v, u)(0, \cdot) = (v_0, u_0)(\cdot),$$

and far-field condition

$$(v, u)(t, x) \xrightarrow{x \rightarrow \pm\infty} (v_\pm, u_\pm). \quad (1.3)$$

System (1.1) was introduced in the context of congested flows, that is in the modeling of flows satisfying the maximal density constraint $\rho = \frac{1}{v} \leq 1$. Equations (1.1) – (1.3) represent an approximation of the following free-congested Navier-Stokes equations

$$\partial_t v - \partial_x u = 0, \quad (1.4a)$$

$$\partial_t u + \partial_x p_\varepsilon(v) - \mu \partial_x \left(\frac{1}{v} \partial_x u \right) = 0, \quad (1.4b)$$

$$v \geq 1, \quad (v-1)p = 0, \quad p \geq 0, \quad (1.4c)$$

with the far-field condition

$$(v, u, p)(t, x) \xrightarrow{x \rightarrow \pm\infty} (v_\pm, u_\pm, p_\pm).$$

One of their main results concerns the existence and qualitative asymptotic behavior of solutions of (1.1)-(1.3). More precisely, we recall the results that interest us. They prove the following proposition.

Proposition 1.1. Assume that the pressure law is given by (1.2).

1. Let $1 < v_- < v_+$ and let u_+, u_- such that

$$(u_+ - u_-)^2 = -(v_+ - v_-)(p_\varepsilon(v_+) - p_\varepsilon(v_-)).$$

Then there exists a unique (up to a shift) traveling front solution of (1.1)-(1.3) $(u, v)(t, x) = (\mathbf{u}_\varepsilon, \mathbf{v}_\varepsilon)(x - s_\varepsilon t)$. The shock speed s_ε satisfies the condition

$$s_\varepsilon^2 = -\frac{p_\varepsilon(v_+) - p_\varepsilon(v_-)}{v_+ - v_-}.$$

2. Take $v_- = 1 + \varepsilon^{\frac{1}{\gamma}}$, $v_+ > 1$ (independent of ε). Let $r = \frac{v_+}{\mu\sqrt{v_+ - 1}}$.

$$\begin{cases} \bar{v}(\xi) = 1 & \text{if } \xi < 0 \\ \bar{v}(\xi) = \frac{v_+}{1 + (v_+ - 1) \exp -r\xi} & \text{if } \xi > 0, \end{cases} \quad \bar{u}'(\xi) = -\frac{u_- - u_+}{v_+ - 1} \bar{v}'$$

is a solution of the limit system (1.4).

These results are proved in Lagrangian coordinates. The aim of this report is to draw inspiration from the proof of this proposition in order to obtain similar results in the Eulerian coordinate setting.

2 Partially congested profiles in Eulerian coordinate

First, we introduce the systems in the Eulerian formulation. We consider the barotropic Navier-Stokes equations, and we refered to that as the singular pressure system

$$\partial_t \rho + \partial_x(\rho u) = 0, \quad (2.1a)$$

$$\partial_t(\rho u) + \partial_x p_\varepsilon(\rho) + \partial_x(\rho u^2) = \mu \partial_x^2 u, \quad (2.1b)$$

where ρ is the density of the fluid, u is the velocity, μ is a viscosity coefficient and p_ε is the pressure given by

$$p_\varepsilon(\rho) = \frac{\varepsilon \rho^\gamma}{(1 - \rho)^\gamma} \quad \gamma > 0, \quad (2.2)$$

and the far-field condition

$$(\rho, u)(t, x) \xrightarrow{x \rightarrow \pm\infty} (\rho_\pm, u_\pm). \quad (2.3)$$

As previously said, it is an approched problem of the free-congested one :

$$\partial_t \rho + \partial_x(\rho u) = 0, \quad (2.4a)$$

$$\partial_t(\rho u) + \partial_x p(\rho) + \partial_x(\rho u^2) = \mu \partial_x^2 u, \quad (2.4b)$$

$$0 \leq \rho \leq 1, \quad (1 - \rho)p = 0, \quad p \geq 0, \quad (2.4c)$$

with the far-field condition

$$(\rho, u, p)(t, x) \xrightarrow{x \rightarrow \pm\infty} (\rho_\pm, u_\pm, p_\pm). \quad (2.5)$$

As said in the introduction, the aim of this paper is to investigate the two systems. First, we focus on system (2.4), and study the existence and uniqueness of a traveling front solution. The second part is devoted to the analysis of system (2.1) where we establish the existence and uniqueness of a traveling front solution. Finally, we show that this solution converges to the solution of the free-congested problem as ε tends to 0.

2.1 Traveling fronts of (2.4).

In this paragraph, we study the existence and properties of traveling fronts of the limit system (2.4). More precisely, we prove the following proposition.

Proposition 2.1. Let $\rho_- = 1 > \rho_+$, $u_- > u_+$. There exist a unique (up to a shift) traveling front solution of (2.4) $(u, \rho, p)(t, x) = (\mathbf{u}, \boldsymbol{\rho}, \mathbf{p})(x - st) = (\mathbf{u}, \boldsymbol{\rho}, \mathbf{p})(\xi)$ satisfying the far field condition

$$(\rho, u, p)(t, x) \xrightarrow{x \rightarrow \pm\infty} (\rho_{\pm}, u_{\pm}, p_{\pm}).$$

This solution is given by :

$$\begin{cases} \rho(\xi) = \rho_- & \text{if } \xi \leq \xi^* \\ \rho(\xi) = \frac{\rho_+}{1 + (\rho_+ - 1)e^{-\frac{\rho_+(s-u_+)}{\mu}(\xi-\xi^*)}} & \text{if } \xi > \xi^* \end{cases} \quad (2.6)$$

$$\begin{cases} \mathbf{u}(\xi) = u_- & \text{if } \xi \leq \xi^* \\ \mathbf{u}(\xi) = (u_- - u_+)e^{-\frac{\rho_+(s-u_+)}{\mu}(\xi-\xi^*)} + u_+ & \text{if } \xi > \xi^* \end{cases} \quad (2.7)$$

$$\begin{cases} \mathbf{p}(\xi) = p_- = \frac{\rho_+(u_+ - u_-)^2}{1 - \rho_+} & \text{if } \xi \leq \xi^* \\ \mathbf{p}(\xi) = 0 & \text{if } \xi > \xi^* \end{cases} \quad (2.8)$$

where the shock speed s verifies

$$s = \frac{\rho_+ u_+ - u_-}{\rho_+ - 1}. \quad (2.9)$$

and p_- is given by

$$p_- = \frac{\rho_+(u_+ - u_-)^2}{1 - \rho_+}. \quad (2.10)$$

Proof. Let (u, ρ, p) be a solution of (2.4) of the form $(\mathbf{u}, \boldsymbol{\rho}, \mathbf{p})(x - st) = (\mathbf{u}, \boldsymbol{\rho}, \mathbf{p})(\xi)$ satisfying the far field condition. Like in [3] it is legitimate to look for a profile which the congested zone is exactly $] - \infty, \xi^*[$ for some $\xi^* \in \mathbb{R}$. It means that in the congested zone (*i.e.* $\xi \in] - \infty, \xi^*[$), $\boldsymbol{\rho} = 1$, $\mathbf{p} > 0$ and in the free zone (*i.e.* $\xi \in]\xi^*, +\infty[$), $\boldsymbol{\rho} < 1$, $\mathbf{p} = 0$. The system (2.4a) - (2.4b) is equivalent to

$$-s\boldsymbol{\rho}' + (\boldsymbol{\rho}\mathbf{u})' = 0, \quad (2.11a)$$

$$-s(\boldsymbol{\rho}\mathbf{u})' + (\boldsymbol{\rho}\mathbf{u}^2)' + \mathbf{p}' = \mu\mathbf{u}''. \quad (2.11b)$$

By plugging (2.11a) in (2.11b), we find $\forall \xi \in \mathbb{R}$

$$-s\boldsymbol{\rho}'(\xi) + (\boldsymbol{\rho}\mathbf{u})'(\xi) = 0, \quad (2.12a)$$

$$-s\boldsymbol{\rho}(\xi)\mathbf{u}'(\xi) + (\boldsymbol{\rho}\mathbf{u})(\xi)\mathbf{u}'(\xi) + \mathbf{p}'(\xi) = \mu\mathbf{u}''(\xi). \quad (2.12b)$$

In the free zone, $\mathbf{p} = 0$, thus by integrating (2.12a) between ξ and $+\infty$, we obtain that

$$\forall \xi > \xi^* \quad s\boldsymbol{\rho}(\xi) = (\boldsymbol{\rho}\mathbf{u})(\xi) + \rho_+(s - u_+). \quad (2.13)$$

By plugging this expression in (2.12b) we find that $\forall \xi > \xi^*$,

$$\mu \mathbf{u}''(\xi) = \rho_+(u_+ - s) \mathbf{u}'(\xi). \quad (2.14)$$

By integration between ξ and $+\infty$, it comes,

$$\mu(u'_+ - \mathbf{u}'(\xi)) = \rho_+(u_+ - s)(u_+ - \mathbf{u}(\xi)).$$

We assume that $\mathbf{u}'$ has a limit at $+\infty$. As $\mathbf{u}$ as a limit at $+\infty$, $u'_+ = 0$. So we obtain :

$$\forall \xi > \xi^*, \quad -\mu \mathbf{u}'(\xi) = \rho_+(u_+ - s)(u_+ - \mathbf{u}(\xi)). \quad (2.15)$$

We find that $\mathbf{u}$ verifies the following differential equation on $]\xi^*, +\infty[$

$$\mathbf{u}' = -\frac{\rho_+(s - u_+)}{\mu} \mathbf{u} - \frac{\rho_+ u_+^2 - s u_+ \rho_+}{\mu}. \quad (2.16)$$

The solution of (2.16) is

$$\mathbf{u}(\xi) = \lambda e^{-\frac{\rho_+(s-u_+)}{\mu}\xi} + u_+ \quad \lambda \in \mathbb{R}. \quad (2.17)$$

We now examine what happens in the congested zone. We know that $\rho = 1$ thus $\rho' = 0$. By (2.11a), it gives $\mathbf{u}' = 0$ and using (2.11b), $p' = 0$ in the congested zone. Thus, $\forall \xi \in]-\infty, \xi^*[$,

$$\begin{cases} \mathbf{u}(\xi) = u_- \\ \rho(\xi) = \rho_- = 1 \\ p(\xi) = p_- \end{cases}$$

As $\mathbf{u}$ is continuous at ξ^* , using (2.17),

$$\mathbf{u}(\xi^*) = \lambda e^{-\frac{\rho_+(s-u_+)}{\mu}\xi^*} + u_+ = u_-$$

which gives

$$\lambda = (u_- - u_+) e^{\frac{\rho_+(s-u_+)}{\mu}\xi^*} > 0.$$

Hence, it gives an expression for $\mathbf{u}$ in the free zone : $\forall \xi > \xi^*$,

$$\mathbf{u}(\xi) = (u_- - u_+) e^{-\frac{\rho_+(s-u_+)}{\mu}(\xi-\xi^*)} + u_+.$$

By using the relation between ρ and $\mathbf{u}$ ((2.13)), we have, $\forall \xi > \xi^*$,

$$\begin{aligned} \rho(\xi) &= \frac{1}{s - \mathbf{u}(\xi)} (s\rho_+ - \rho_+ u_+) \\ &= \frac{(\rho_+ s - \rho_+ u_+)}{s - (u_- - u_+) e^{-\frac{\rho_+(s-u_+)}{\mu}(\xi-\xi^*)} + u_+}. \end{aligned}$$

Therefore, we have obtained expressions for ρ and $\mathbf{u}$. We now express p_- and s in terms of the problem data. The latter will in particular allow us to derive a simpler expression

for ρ .

By integrating (2.11a) on $\mathbb{R}$, it comes

$$-s(\rho_+ - 1) + (\rho_+ u_+ - u_-) = 0$$

which yields

$$s = \frac{\rho_+ u_+ - u_-}{\rho_+ - 1}. \quad (2.18)$$

By assuming the continuity of $p - \mu \mathbf{u}'$ at $\xi = \xi^*$,

$$\begin{aligned} \lim_{\xi \rightarrow \xi_-^*} [p(\xi) - \mu \mathbf{u}'(\xi)] &= \lim_{\xi \rightarrow \xi_+^*} [p(\xi) - \mu \mathbf{u}'(\xi)] \\ \Leftrightarrow p_- - \mu \lim_{\xi \rightarrow \xi_-^*} \mathbf{u}'(\xi) &= 0 - \mu \lim_{\xi \rightarrow \xi_+^*} \mathbf{u}'(\xi) \\ \Leftrightarrow p_- &= -\mu \lim_{\xi \rightarrow \xi_+^*} \mathbf{u}'(\xi). \end{aligned}$$

Taking the limit as $\xi \rightarrow \xi_+^*$ in (2.15) and using the expression of s in (2.18), it appears

$$\begin{aligned} p_- &= \rho_+ (u_+ - s)(u_+ - u_-) \\ &= \rho_+ \left(\frac{u_+ (\rho_+ - 1) - \rho_+ u_+ + u_-}{\rho_+ - 1} \right) (u_+ - u_-) \end{aligned}$$

and thus

$$p_- = \frac{\rho_+ (u_+ - u_-)^2}{1 - \rho_+}. \quad (2.19)$$

Using (2.18) we find another expression for ρ in the free zone : $\forall \xi > \xi^*$,

$$\rho(\xi) = \frac{\rho_+}{1 + (\rho_+ - 1) e^{-\frac{\rho_+ (s - u_+)}{\mu} (\xi - \xi^*)}}.$$

To conclude, we have obtained that if (u, ρ, p) be a solution of (2.4) of the form $(\mathbf{u}, \boldsymbol{\rho}, \mathbf{p})(x - st)$ (with a congested zone given by $] - \infty, \xi^* [$) satisfying the far field condition

$$(\rho, u, p)(t, x) \xrightarrow{x \rightarrow \pm \infty} (\rho_{\pm}, u_{\pm}, p_{\pm}),$$

$(\mathbf{u}, \boldsymbol{\rho}, \mathbf{p})$ must be given by

$$\begin{cases} \rho(\xi) = \rho_- & \text{if } \xi \leq \xi^* \\ \rho(\xi) = \frac{\rho_+}{1 + (\rho_+ - 1) e^{-\frac{\rho_+ (s - u_+)}{\mu} (\xi - \xi^*)}} & \text{if } \xi > \xi^* \end{cases} \quad (2.20)$$

$$\begin{cases} \mathbf{u}(\xi) = u_- & \text{if } \xi \leq \xi^* \\ \mathbf{u}(\xi) = (u_- - u_+) e^{-\frac{\rho_+ (s - u_+)}{\mu} (\xi - \xi^*)} + u_+ & \text{if } \xi > \xi^* \end{cases} \quad (2.21)$$

$$\begin{cases} \mathbf{p}(\xi) = p_- = \frac{\rho_+ (u_+ - u_-)^2}{1 - \rho_+} & \text{if } \xi \leq \xi^* \\ \mathbf{p}(\xi) = 0 & \text{if } \xi > \xi^* \end{cases} \quad (2.22)$$

Conversely, by taking the limit in the equations above, we find that (u, ρ, p) are satisfying the far field conditions. Moreover, it can be verified that (u, ρ, p) are solutions of (2.4). Therefore there exist a unique solution of (2.4) if we fixed the congested zone as $] - \infty, \xi^* [$. Classically, all traveling waves solutions of (2.4) satisfying (2.5) are translation of (2.20)-(2.22). $\square$

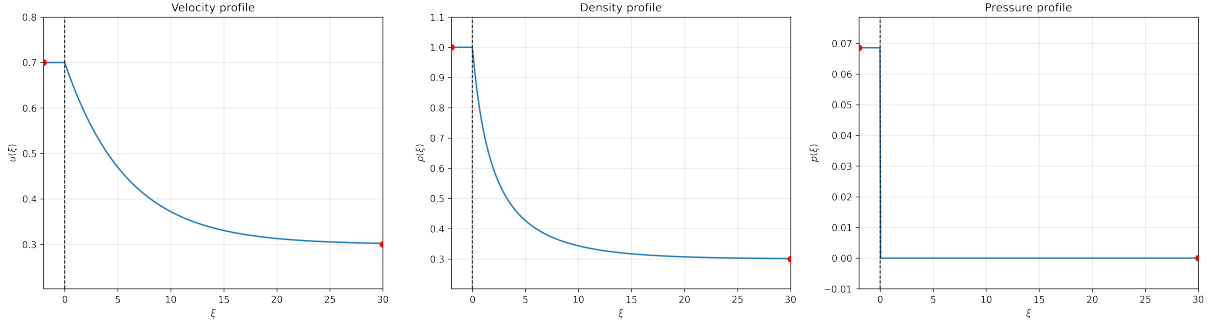


Figure 2.1: Behavior of the solution (2.20)-(2.22) for $\rho_+ = 0.3$, $u_- = 0.7$, $u_+ = 0.3$

Remark. To find an expression for ρ and u , we first obtained an ODE on u from which we deduced the expression of ρ . However, we could have find an ODE on ρ instead. Indeed by plugging (2.16) in (2.11a), we find that in the free zone, ρ satisfies

$$\rho' = (s - u_+) \frac{\rho}{\mu} (\rho_+ - \rho). \quad (2.23)$$

We will see that a parallel can be drawn with this equation later.

To conclude this paragraph, we discuss the behaviour of the speed u and the density ρ in the free zone.

Lemma 2.1. u and ρ are strictly decreasing in the free zone.

Proof. We differentiate u in the free zone, by -(2.21)- it comes,

$$u'(\xi) = -(u_- - u_+) \frac{\rho_+(s - u_+)}{\mu} e^{-\frac{\rho_+(s - u_+)}{\mu}(\xi - \xi^*)}.$$

$u_- - u_+ > 0$, $\rho_+ > 0$ and $s - u_+ = \frac{u_+ - u_-}{\rho_+ - 1} > 0$ thus $u'(\xi) < 0$. Therefore, u is strictly decreasing in the free zone. We differentiate ρ in the free zone, by (2.20) it comes,

$$\rho'(\xi) = \rho_+ \frac{(\rho_+ - 1) \frac{\rho_+(s - u_+)}{\mu} e^{-\frac{\rho_+(s - u_+)}{\mu}(\xi - \xi^*)}}{\left(1 + (\rho_+ - 1) e^{-\frac{\rho_+(s - u_+)}{\mu}(\xi - \xi^*)}\right)^2}.$$

$\rho_+ - 1 < 0$, $\rho_+ > 0$ and $\frac{\rho_+(s - u_+)}{\mu} > 0$, thus $\rho'(\xi) < 0$. Therefore, ρ is strictly decreasing in the free zone. $\square$

2.2 Existence and uniqueness (up to a shift) of traveling fronts of (2.1).

In this paragraph, we focus on the system (2.1). We prove the following proposition.

Proposition 2.2. Assume that the pressure law is given by (2.2). Let $0 < \rho_+ < \rho_- < 1$ and u_+, u_- such that

$$(\rho_- u_- - \rho_+ u_+)^2 = (\rho_- - \rho_+)(p_\varepsilon(\rho_-) - p_\varepsilon(\rho_+) + \rho_- u_-^2 - \rho_+ u_+^2)$$

. Then there exists a unique (up to a shift) traveling front solution of (2.1)-(2.3) $(\rho, u)(t, x) = (\boldsymbol{\rho}_\varepsilon, \mathbf{u}_\varepsilon)(x - s_\varepsilon t)$ with $\mathbf{u}'_\varepsilon$ vanished at infinity. The shock speed s_ε satisfies the *Rankine-Hugoniot* condition

$$s_\varepsilon^2 = \frac{p_\varepsilon(\rho_-) - p_\varepsilon(\rho_+) + \rho_- u_-^2 - \rho_+ u_+^2}{\rho_- - \rho_+}.$$

Proof. Assume that (ρ, u) is a solution of the form $(\boldsymbol{\rho}_\varepsilon, \mathbf{u}_\varepsilon)(x - s_\varepsilon t)$. Plugging this expression in (2.4), by setting $\xi = x - s_\varepsilon t$ we find

$$-s_\varepsilon \boldsymbol{\rho}'_\varepsilon(\xi) + (\boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon)'(\xi) = 0, \quad (2.24a)$$

$$-s_\varepsilon (\boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon)'(\xi) + (p_\varepsilon(\boldsymbol{\rho}_\varepsilon))'(\xi) + (\boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon^2)'(\xi) = \mu \mathbf{u}''_\varepsilon. \quad (2.24b)$$

We integrate the previous system on $] -\infty, \xi[$:

$$\begin{cases} -s_\varepsilon(\boldsymbol{\rho}_\varepsilon - \rho_-) + \boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon - \rho_- u_- = 0, \\ -s_\varepsilon(\boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon - \rho_- u_-) + p_\varepsilon(\boldsymbol{\rho}_\varepsilon) - p_\varepsilon(\rho_-) + \boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon^2 - \rho_- u_-^2 = \mu \mathbf{u}'_\varepsilon. \end{cases}$$

And now the case for $]\xi, +\infty[$:

$$\begin{cases} -s_\varepsilon(\rho_+ - \boldsymbol{\rho}_\varepsilon) + \rho_+ u_+ - \boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon = 0, \\ -s_\varepsilon(\rho_+ u_+ - \boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon) + p_\varepsilon(\rho_+) - p_\varepsilon(\boldsymbol{\rho}_\varepsilon) + \rho_+ u_+^2 - \boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon^2 = -\mu \mathbf{u}'_\varepsilon. \end{cases}$$

We sum up those informations in the following system,

$$s_\varepsilon \boldsymbol{\rho}_\varepsilon - \boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon = s_\varepsilon \rho_\pm - \rho_\pm u_\pm, \quad (2.25a)$$

$$-s_\varepsilon \boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon + p_\varepsilon(\boldsymbol{\rho}_\varepsilon) - \mu \mathbf{u}'_\varepsilon + \boldsymbol{\rho}_\varepsilon \mathbf{u}_\varepsilon^2 = -s_\varepsilon \rho_\pm u_\pm + p_\varepsilon(\rho_\pm) + \rho_\pm u_\pm^2. \quad (2.25b)$$

If we sum the first equations in the two first integrated system, we obtain that the shock speed verify :

$$s_\varepsilon = \frac{\rho_- u_- - \rho_+ u_+}{\rho_- - \rho_+}. \quad (2.26)$$

In the same way regarding to the second equation it comes :

$$s_\varepsilon = \frac{p_\varepsilon(\rho_-) - p_\varepsilon(\rho_+) + \rho_- u_-^2 - \rho_+ u_+^2}{\rho_- u_- - \rho_+ u_+}. \quad (2.27)$$

This leads to the condition :

$$(\rho_- u_- - \rho_+ u_+)^2 = (\rho_- - \rho_+)(p_\varepsilon(\rho_-) - p_\varepsilon(\rho_+) + \rho_- u_-^2 - \rho_+ u_+^2). \quad (2.28)$$

By plugging (2.28) in (2.26), we obtain :

$$s_\varepsilon = \pm \sqrt{-\frac{p_\varepsilon(\rho_+) - p_\varepsilon(\rho_-) + \rho_+ u_+^2 - \rho_- u_-^2}{\rho_- - \rho_+}}. \quad (2.29)$$

Now, as [3] suggest, we need to have an ODE for $\boldsymbol{\rho}_\varepsilon$ or for $\mathbf{u}_\varepsilon$. Here, we obtain an ODE for $\mathbf{u}_\varepsilon$ since it will make it easier to deduce the existence and uniqueness of a traveling

front solution. Later on, we will obtain an ODE for ρ_ε which will be useful in the sequel. We can rewrite (2.24a) as

$$(\rho_\varepsilon(\mathbf{u}_\varepsilon - s_\varepsilon))'(\xi) = 0.$$

Let $q_\varepsilon = \rho_\varepsilon(\mathbf{u}_\varepsilon - s_\varepsilon)$. Thus, $q'_\varepsilon = 0$ on $\mathbb{R}$. Furthermore, (2.24b) can be rewritten as

$$\mathbf{u}'_\varepsilon(\rho_\varepsilon \mathbf{u}_\varepsilon - s_\varepsilon \rho_\varepsilon) + p'_\varepsilon(\rho_\varepsilon) = \mu \mathbf{u}''_\varepsilon$$

by using (2.24a). The definition of q_ε yields

$$\mathbf{u}'_\varepsilon q_\varepsilon + p'_\varepsilon(\rho_\varepsilon) = \mu \mathbf{u}''_\varepsilon.$$

By integrating this expression on $] - \infty, \xi[$ and on $] \xi, +\infty[$, we find,

$$(\mathbf{u}_\varepsilon - u_-)q_\varepsilon + (p_\varepsilon(\rho_\varepsilon) - p_\varepsilon(\rho_-)) = \mu(\mathbf{u}'_\varepsilon - u'_-) = \mu \mathbf{u}'_\varepsilon, \quad (2.30a)$$

$$(u_+ - \mathbf{u}_\varepsilon)q_\varepsilon + (p_\varepsilon(\rho_+) - p_\varepsilon(\rho_\varepsilon)) = -\mu \mathbf{u}'_\varepsilon, \quad (2.30b)$$

This leads to

$$(u_+ - u_-)q_\varepsilon + p_\varepsilon(\rho_+) - p_\varepsilon(\rho_-) = 0,$$

thus

$$q_\varepsilon = \frac{p_\varepsilon(\rho_-) - p_\varepsilon(\rho_+)}{u_+ - u_-}.$$

By plugging the expression of q_ε in (2.30a) and (2.30b), it comes,

$$\begin{aligned} \mathbf{u}'_\varepsilon &= \frac{1}{\mu}(\mathbf{u}_\varepsilon - u_-) \left(\frac{p_\varepsilon(\rho_-) - p_\varepsilon(\rho_+)}{u_+ - u_-} + \frac{(p_\varepsilon(\rho_\varepsilon) - p_\varepsilon(\rho_-))}{(\mathbf{u}_\varepsilon - u_-)} \right) \\ &= \frac{1}{\mu}(\mathbf{u}_\varepsilon - u_+) \left(\frac{p_\varepsilon(\rho_-) - p_\varepsilon(\rho_+)}{u_+ - u_-} + \frac{(p_\varepsilon(\rho_\varepsilon) - p_\varepsilon(\rho_+))}{(\mathbf{u}_\varepsilon - u_+)} \right). \end{aligned}$$

This expression still contains terms in ρ_ε . As $\rho_\varepsilon = \frac{q_\varepsilon}{\mathbf{u}_\varepsilon - s_\varepsilon}$, we set

$$\tilde{p}_\varepsilon(\mathbf{u}_\varepsilon) = p_\varepsilon(\rho_\varepsilon) = p_\varepsilon\left(\frac{q_\varepsilon}{\mathbf{u}_\varepsilon - s_\varepsilon}\right).$$

This yields an ODE for $\mathbf{u}_\varepsilon$,

$$\begin{aligned} \mathbf{u}'_\varepsilon &= \frac{1}{\mu}(\mathbf{u}_\varepsilon - u_-) \left(-\frac{\tilde{p}_\varepsilon(u_+) - \tilde{p}_\varepsilon(u_-)}{u_+ - u_-} + \frac{\tilde{p}_\varepsilon(\mathbf{u}_\varepsilon) - \tilde{p}_\varepsilon(u_-)}{(\mathbf{u}_\varepsilon - u_-)} \right) \\ &= \frac{1}{\mu}(\mathbf{u}_\varepsilon - u_+) \left(-\frac{\tilde{p}_\varepsilon(u_+) - \tilde{p}_\varepsilon(u_-)}{u_+ - u_-} + \frac{\tilde{p}_\varepsilon(\mathbf{u}_\varepsilon) - \tilde{p}_\varepsilon(u_+)}{(\mathbf{u}_\varepsilon - u_+)} \right). \end{aligned} \quad (2.31)$$

Now, assume that $u_- > u_+$, and let $u_0 \in]u_+, u_-[$ be arbitrary. We consider the Cauchy problem (2.31) with initial data $\mathbf{u}_\varepsilon(0) = u_0$. p_ε is of class C^1 hence $\tilde{p}_\varepsilon$ as well. By the Cauchy–Lipschitz theorem, there exists a unique maximal solution $\mathbf{u}_\varepsilon$ to this problem. Moreover, it is straightforward to verify that $\mathbf{u} = u_\pm$ are constant solutions of (2.31). This implies that $\mathbf{u}_\varepsilon(\xi) \in]u_+, u_-[$ for all ξ in the interval of existence, and therefore the solution is global. It remains to verify the far-field condition in order to ensure that the obtained solution is consistent with (2.1)–(2.3).

Lemma 2.2. $\mathbf{u}_\varepsilon$ is strictly decreasing on $\mathbb{R}$. More precisely, $\mathbf{u}_\varepsilon : \mathbb{R} \rightarrow]u_+, u_-[$ is one-to-one and onto.

Proof. Note that $\tilde{p}_\varepsilon$ is of class C^∞ on $]u_+, u_-[$ and $\forall u \in]u_+, u_-[$,

$$\tilde{p}'_\varepsilon(u) = \frac{-q_\varepsilon}{(u - s_\varepsilon)^2} p'_\varepsilon\left(\frac{q_\varepsilon}{u - s_\varepsilon}\right).$$

And, $q_\varepsilon = \frac{p_\varepsilon(\rho_-) - p_\varepsilon(\rho_+)}{u_+ - u_-} < 0$ since p_ε is increasing. Hence $\tilde{p}_\varepsilon < 0$ so $\tilde{p}_\varepsilon$ is strictly increasing. Moreover, $\forall u \in]u_+, u_-[$,

$$\tilde{p}''_\varepsilon(u) = \frac{2\rho_\varepsilon}{(u - s_\varepsilon)^2} p'_\varepsilon\left(\frac{q_\varepsilon}{u - s_\varepsilon}\right) + \frac{q_\varepsilon^2}{(u - s_\varepsilon)^4} p''_\varepsilon\left(\frac{q_\varepsilon}{u - s_\varepsilon}\right).$$

This quantity is strictly positive, since p_ε is strictly increasing and strictly convex, and $\tilde{p}_\varepsilon$ is consequently strictly convex. Hence,

$$\frac{\tilde{p}_\varepsilon(u_+) - \tilde{p}_\varepsilon(u_-)}{u_+ - u_-} < \frac{\tilde{p}_\varepsilon(\mathbf{u}_\varepsilon) - \tilde{p}_\varepsilon(u_-)}{(\mathbf{u}_\varepsilon - u_-)}$$

and since these quantities are positive,

$$-\frac{\tilde{p}_\varepsilon(u_+) - \tilde{p}_\varepsilon(u_-)}{u_+ - u_-} + \frac{\tilde{p}_\varepsilon(\mathbf{u}_\varepsilon) - \tilde{p}_\varepsilon(u_-)}{(\mathbf{u}_\varepsilon - u_-)} > 0. \quad (2.32)$$

Hence, we deduce that $\mathbf{u}'_\varepsilon < 0$, therefore, $\mathbf{u}_\varepsilon$ is strictly decreasing on $\mathbb{R}$. Moreover, $\mathbf{u}_\varepsilon(\xi)$ as a limit as $\xi \rightarrow +\infty$ and $\xi \rightarrow -\infty$ which we denote by $\mathbf{u}_\varepsilon^-$ and $\mathbf{u}_\varepsilon^+$, $\mathbf{u}_\varepsilon$ is of class C^1 , $\mathbf{u}'_\varepsilon$ as a limit, thus, $\mathbf{u}'_\varepsilon$ tends to 0 at $+\infty$ and $-\infty$. From (2.32), the strict monotonicity of $\mathbf{u}_\varepsilon$ and (2.31), we deduce that, $\mathbf{u}_\varepsilon^+ = u_+$ and $\mathbf{u}_\varepsilon^- = u_-$. $\square$

Finally, if (ρ, u) is a solution of (2.1)-(2.3) of the form $(\rho_\varepsilon, \mathbf{u}_\varepsilon)(x - s_\varepsilon t)$, $\mathbf{u}_\varepsilon$ is unique and using (2.25), we deduce that ρ_ε is unique, which yields the uniqueness of (ρ, u) .

For the existence, let $\mathbf{u}_\varepsilon$ be the solution of (2.31). We saw that $\mathbf{u}_\varepsilon^+ = u_+$ and $\mathbf{u}_\varepsilon^- = u_-$. In addition, if $(\rho_\varepsilon, \mathbf{u}_\varepsilon)$ is a solution of (2.1)-(2.3), ρ_ε must satisfy (2.25a)-(2.25b). Defined by (2.1)-(2.3), ρ_ε as a limit at $+\infty$ and $-\infty$. Taking the limit in (2.25a) it comes $\rho_\varepsilon^+ = \rho_+$ and $\rho_\varepsilon^- = \rho_-$. Finally, by differentiating the equations (2.25a)-(2.25b), we find, $\forall \xi \in \mathbb{R}$,

$$\begin{aligned} \partial_t \rho_\varepsilon(\xi) + \partial_x(\rho_\varepsilon \mathbf{u}_\varepsilon)(\xi) &= 0, \\ \partial_t(\rho_\varepsilon \mathbf{u}_\varepsilon)(\xi) + \partial_x p_\varepsilon(\rho_\varepsilon)(\xi) + \partial_x(\rho_\varepsilon \mathbf{u}_\varepsilon^2)(\xi) &= \mu \partial_x^2 \mathbf{u}_\varepsilon(\xi). \end{aligned}$$

Classically, all other solutions of (2.31) satisfying the far field condition (2.3) are translation of this profile. This proves the proposition. $\square$

Remark. We have found an ODE for $\mathbf{u}_\varepsilon$ which was adapted to the proof we made. We can obtain an ODE on ρ_ε which has the advantage to making the speed s_ε appear, and we will base the next section on this equation. This ODE is the following :

$$\begin{aligned} \rho'_\varepsilon &= \frac{\rho_\varepsilon}{\mu(s_\varepsilon - u_+)} \left((s_\varepsilon - u_+)^2(\rho_+ - \rho_\varepsilon) + \frac{\rho_\varepsilon}{\rho_+} (p_\varepsilon(\rho_\varepsilon) - p_\varepsilon(\rho_+)) \right) \\ &= \frac{\rho_\varepsilon}{\mu(s_\varepsilon - u_-)} \left((s_\varepsilon - u_-)^2(\rho_- - \rho_\varepsilon) + \frac{\rho_\varepsilon}{\rho_-} (p_\varepsilon(\rho_\varepsilon) - p_\varepsilon(\rho_-)) \right). \end{aligned} \quad (2.33)$$

We refer to the appendix for details on how this equation is obtained.

2.3 Qualitative asymptotic description of traveling fronts.

In this section we are interested in the case when $0 < \rho_+ < 1$ is a fixed number, independent of ε . We investigate the asymptotic behavior of traveling fronts for the system (2.1).

We focus on traveling fronts such that $s_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} \bar{s}$ when $\varepsilon \rightarrow 0$. Let $\rho_\varepsilon^- = \frac{1}{1 + \varepsilon^{\frac{1}{\gamma}}}$. Since

$$s_\varepsilon = \frac{\rho_\varepsilon^- u_\varepsilon^- - \rho_+ u_+}{\rho_- - \rho_+},$$

u_ε^- has a limite as $\varepsilon \rightarrow 0$ denoted by u_- . Since $\rho_\varepsilon^- \xrightarrow{\varepsilon \rightarrow 0} 1$, we find

$$s_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} \frac{u_- - \rho_+ u_+}{1 - \rho_+} = \bar{s}. \quad (2.34)$$

Remark. $\bar{s}$ equals the speed of the traveling wave in the free-congested system.

Proposition 2.3. There exist $\bar{\rho}, \bar{u} \in W^{1,\infty}(\mathbb{R})$ such that up to a subsequence, $\rho_\varepsilon \rightarrow \bar{\rho}$ and $u_\varepsilon \rightarrow \bar{u}$ pointwise on $\mathbb{R}$. And, $\forall \xi \in \mathbb{R}$, $p_\varepsilon(\rho_\varepsilon)(\xi)$ has a limit as $\varepsilon \rightarrow 0$ which we denote $\bar{p}(\xi)$. Moreover $(\bar{\rho}, \bar{u}, \bar{p})$, is the solution of (2.4) that verifies the far field condition, and where the congested domain is $] - \infty, \bar{\xi}[$ where $\bar{\xi} = \frac{\ln(\rho_+ + 1)}{\frac{\rho_+(\bar{s} - u_+)}{\mu}}$.

Proof. In order to fix the shift, let us consider the solution of (2.33) with $\rho_\varepsilon(0) = \frac{1 + \rho_+}{2}$. Let $\varepsilon > 0$. We have $p_\varepsilon(\rho_\varepsilon^-) = \frac{\varepsilon}{(1 + \varepsilon^{\frac{1}{\gamma}} - 1)^\gamma} = 1$. Now, $\forall \xi \in \mathbb{R}$, $\rho_\varepsilon^- > \rho_\varepsilon(\xi)$, thus $\forall \xi \in \mathbb{R}$, $p_\varepsilon(\rho_\varepsilon(\xi)) \leq 1$, using the fact that p_ε is increasing. Hence, by (2.33), $\forall \xi \in \mathbb{R}$,

$$\begin{aligned} |\rho'_\varepsilon(\xi)| &\leq \frac{\rho_\varepsilon(\xi)}{\mu |s_\varepsilon - u_+|} \left((s_\varepsilon - u_+)^2 |\rho_+ - \rho_\varepsilon(\xi)| + \frac{\rho_\varepsilon(\xi)}{\rho_+} |p_\varepsilon(\rho_\varepsilon) - p_\varepsilon(\rho_+)| \right) \\ &\leq \frac{\rho_\varepsilon(\xi)}{\mu |s_\varepsilon - u_+|} \left((|s_\varepsilon|^2 + |u_+|^2)(\rho_+ + 1) + \frac{2}{\rho_+} \right). \end{aligned}$$

Moreover, s_ε converges so is bounded. Hence, there exist $M > 0$, independant of ε such that $\forall \xi \in \mathbb{R}$,

$$|\rho'_\varepsilon(\xi)| \leq M.$$

Thus the family $(\rho_\varepsilon)_\varepsilon$ is equi-Lipschitz hence equicontinuous.

In addition, according to the previous paragraph, $\rho_\varepsilon(\xi) \in]0, 1[$, $\forall \xi \in \mathbb{R}$. Therefore, using Ascoli theorem, $(\rho_\varepsilon)_\varepsilon$ is relatively compact in $(C^0([-R, R], \mathbb{R}), \|\cdot\|_\infty)$, for all $R > 0$. Moreover, using the fact that a pointwise limit of Lipschitz functions is itself Lipschitz, there exists $\bar{\rho} \in W^{1,\infty}(\mathbb{R})$ such that up to a subsequence,

$$\rho_\varepsilon \xrightarrow{\varepsilon \rightarrow 0} \bar{\rho} \text{ in } (C^0([-R, R], \mathbb{R}), \|\cdot\|_\infty) \quad \forall R > 0.$$

Since ρ_ε is decreasing, so is $\bar{\rho}$. And $\bar{\rho} \in [\rho_+, 1]$, and by pointwise convergence, $\bar{\rho}(0) = \frac{\rho_+ + 1}{2}$. We define

$$\bar{\xi} = \inf\{\xi \in \mathbb{R}, \bar{\rho} < 1\} \in] - \infty, 0[.$$

thus $\bar{\rho}(\xi) = 0 \ \forall \xi \in]-\infty, \bar{\xi}[$. Since $\forall \xi > \bar{\xi}$, $\bar{\rho}(\xi) < 1$, and $\rho_\varepsilon(\xi) \xrightarrow{\varepsilon \rightarrow 0} \bar{\rho}(\xi)$, we find,

$$p_\varepsilon(\rho_\varepsilon(\xi)) = \frac{\varepsilon \rho_\varepsilon^\gamma(\xi)}{(1 - \rho_\varepsilon(\xi))^\gamma} \xrightarrow{\varepsilon \rightarrow 0} 0.$$

Using the fact that ρ_ε pointwise converges to $\bar{\rho}$ on $\mathbb{R}$, we can pass to the limit in (2.33). In the limit as $\varepsilon \rightarrow 0$, we obtain that on $]\bar{\xi}, +\infty[$, $\bar{\rho}$ is a solution of the logistic equation

$$\bar{\rho}' = \frac{\bar{\rho}}{\mu(\bar{s} - u_+)}(\bar{s} - u_+)^2(\rho_+ - \bar{\rho}) \quad (2.35)$$

$$= \frac{(\bar{s} - u_+)\bar{\rho}}{\mu}(\rho_+ - \bar{\rho}). \quad (2.36)$$

Remark. This relation should be compared with (2.23): the limit profile satisfies the same differential equation in the free zone as the solution of the free congested system.

If we set $\bar{\rho}(0) = \frac{1 + \rho_+}{2}$, it comes,

$$\begin{cases} \bar{\rho}(\xi) = 1 & \text{if } \xi \leq \bar{\xi} \\ \bar{\rho}(\xi) = \frac{\rho_+}{1 + \frac{\rho_+ - 1}{\rho_+ + 1} e^{-\frac{\rho_+(\bar{s} - u_+)}{\mu}\xi}} & \text{if } \xi > \bar{\xi} \end{cases} \quad (2.37)$$

This allows us to find an explicit expression for $\bar{\xi}$, namely

$$\bar{\xi} = \frac{\ln(\rho_+ + 1)}{\frac{\rho_+(\bar{s} - u_+)}{\mu}}.$$

By (2.25a),

$$\mathbf{u}_\varepsilon = s_\varepsilon + \frac{\rho_+ u_+ - s_\varepsilon \rho_+}{\rho_\varepsilon}$$

therefore $\mathbf{u}_\varepsilon$ has a limit as ε tends to 0, denoted by $\bar{\mathbf{u}}$, and

$$\bar{\mathbf{u}} = \bar{s} + \frac{\rho_+ u_+ - \bar{s} \rho_+}{\bar{\rho}}. \quad (2.38)$$

Thus, if $\xi \leq \bar{\xi}$, by (2.34)

$$\bar{\mathbf{u}}(\xi) = \rho_+ u_+ - \bar{s} \rho_+ = u_-.$$

And by (2.37) and (2.38), $\forall \xi > \bar{\xi}$,

$$\begin{aligned} \bar{\mathbf{u}}(\xi) &= \bar{s} + \frac{(\rho_+ u_+ - \bar{s} \rho_+) \left(1 + \frac{\rho_+ - 1}{\rho_+ + 1} e^{-\frac{\rho_+(\bar{s} - u_+)}{\mu}\xi}\right)}{\rho_+} \\ &= \bar{s} + (u_+ - \bar{s}) \left(1 + \frac{\rho_+ - 1}{\rho_+ + 1} e^{-\frac{\rho_+(\bar{s} - u_+)}{\mu}\xi}\right) \\ &= u_+ + (u_+ - \bar{s}) \frac{\rho_+ - 1}{\rho_+ + 1} e^{-\frac{\rho_+(\bar{s} - u_+)}{\mu}\xi} \\ &= u_+ + \frac{u_- - u_+}{\rho_+ + 1} e^{-\frac{\rho_+(\bar{s} - u_+)}{\mu}\xi}. \end{aligned}$$

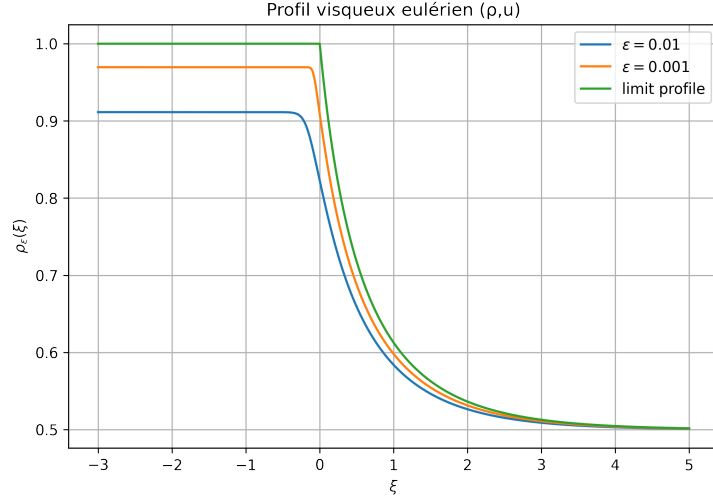


Figure 2.2: Asymptotic behavior of the profiles ρ_ε with $\rho_-^\varepsilon = \frac{1}{1+\varepsilon^{\frac{1}{\gamma}}}$, $\rho_+ = 0.5$ and $\gamma = 2$.

The shift is fixed by prescribing $\rho_\varepsilon(0) = \frac{1}{1 + \varepsilon^{\frac{1}{\gamma+1}}}$

Hence,

$$\begin{cases} \bar{u}(\xi) = 1 & \text{if } \xi \leq \bar{\xi} \\ \bar{u}(\xi) = u_+ + \frac{u_- - u_+}{\rho_+ + 1} e^{-\frac{\rho_+ (\bar{s} - u_+)}{\mu} \xi} & \text{if } \xi > \bar{\xi} \end{cases} \quad (2.39)$$

In addition, at the limit in (2.33), since $\rho_\varepsilon^- = 1$ and $\bar{\rho}(\xi) = 1 \forall \xi \leq \bar{\xi}$, we find we find that $\bar{p}(\xi) = 1, \forall \xi \leq \bar{\xi}$. And, as mentioned earlier, $\bar{p}(\xi) = 0 \forall \xi > \bar{\xi}$.

Thus, we have $0 \leq \bar{\rho} \leq 1$, $(1 - \bar{\rho})\bar{p} = 0$, $\bar{p} \geq 0$, and it can be verified that $\bar{\rho}, \bar{u}, \bar{p}$ are indeed solutions of the free-congested system. $\square$

Remark. In this case, as opposed to the free congested case, we first set an initial condition and then determined $\bar{\xi}$. If we set $\xi^* = \bar{\xi} = \frac{\ln(\rho_+ + 1)}{\frac{\rho_+ (\bar{s} - u_+)}{\mu}}$ in (2.20), this yields the same solution as in (2.37).

Appendix

A Eulerian to Lagrangian coordinate system

This section is devoted to the formalization of the change of coordinates. We start from the one-dimensional Navier–Stokes system written in Eulerian coordinates (2.4). We introduce the flow map $X \mapsto x(t, X)$, defined as the maximal solution of

$$\begin{cases} \partial_t x(t, X) = u(t, x(t, X)), \\ x(0, X) = X. \end{cases} \quad (\text{A.1})$$

We define

$$v(t, X) = \frac{1}{\rho(t, x(t, X))} = \partial_X x(t, X), \quad U(t, X) = u(t, x(t, X)), \quad P(t, X) = p(t, x(t, X)).$$

With this notation, we obtain

$$\partial_t v = \partial_t(\partial_X x) = \partial_X(\partial_t x) = \partial_X U,$$

which yields

$$\partial_t v - \partial_X U = 0.$$

Moreover, equation (2.4b) can be rewritten as

$$\begin{aligned} \partial_t(\rho u) + \partial_x p + \partial_x(\rho u^2) &= \mu \partial_x^2 u \Leftrightarrow (\partial_t \rho)u + \rho(\partial_t u) + \partial_x p - \mu \partial_x^2 u + \partial_x(\rho u^2) = 0 \\ &\Leftrightarrow \rho(\partial_t u + u \partial_x u) + \partial_x p - \mu \partial_x^2 u = 0. \end{aligned}$$

Dividing by ρ leads to

$$\partial_t u + u \partial_x u + \frac{1}{\rho} \partial_x p - \frac{\mu}{\rho} \partial_x^2 u = 0. \quad (\text{A.2})$$

We now express each term in Lagrangian coordinates:

- $\partial_t U = \partial_t u + u \partial_x u$,
- $\partial_X P = \partial_X x \partial_x p = v \partial_x p$,
- $\partial_X U = \partial_X(\partial_t x) = \partial_X u = \partial_X x \partial_x u = v \partial_x u$,
- $\partial_x^2 u = \partial_x \left(\frac{1}{v} \partial_X U \right) = \frac{1}{v} \partial_X \left(\frac{1}{v} \partial_X U \right)$, and therefore

$$\frac{\mu}{\rho} \partial_x^2 u = \mu v \frac{1}{v} \partial_X \left(\frac{1}{v} \partial_X U \right) = \mu \partial_X \left(\frac{1}{v} \partial_X U \right).$$

Finally, equation (A.2) can be rewritten in Lagrangian coordinates as

$$\partial_t U + \partial_X P - \mu \partial_X \left(\frac{1}{v} \partial_X U \right) = 0.$$

By collecting all these results, we obtain the system (1.4).

B ODE on ρ_ε for the singular pressure system (2.1)

The equation (2.24a) gives $-s_\varepsilon \rho'_\varepsilon + \rho'_\varepsilon \mathbf{u}_\varepsilon + \mathbf{u}'_\varepsilon \rho_\varepsilon = 0$ which can be rewritten as (under $\rho_\varepsilon \neq 0$) :

$$\mathbf{u}'_\varepsilon = \frac{\rho'_\varepsilon}{\rho_\varepsilon} (s_\varepsilon - \mathbf{u}_\varepsilon) \quad (\text{B.1})$$

Using (B.1) in (2.25b),

$$\begin{aligned} & -s_\varepsilon \rho_\varepsilon \mathbf{u}_\varepsilon + p_\varepsilon(\rho_\varepsilon) - \mu \frac{\rho'_\varepsilon}{\rho_\varepsilon} (s_\varepsilon - \mathbf{u}_\varepsilon) + \rho_\varepsilon \mathbf{u}_\varepsilon^2 = -s_\varepsilon \rho_\pm u_\pm + p_\varepsilon(\rho_\pm) + \rho_\pm u_\pm^2 \\ & \stackrel{(2.25a)}{\Leftrightarrow} -s_\varepsilon^2 (\rho_\varepsilon - \rho_\pm) - s_\varepsilon \rho_\pm u_\pm + p_\varepsilon(\rho_\varepsilon) - \mu \frac{\rho'_\varepsilon}{\rho_\varepsilon} \left(s_\varepsilon - \frac{s_\varepsilon (\rho_\varepsilon - \rho_\pm) + \rho_\pm u_\pm}{\rho_\varepsilon} \right) + \\ & \quad \rho_\varepsilon \left(\frac{s_\varepsilon (\rho_\varepsilon - \rho_\pm) + \rho_\pm u_\pm}{\rho_\varepsilon} \right)^2 = -s_\varepsilon \rho_\pm u_\pm + p_\varepsilon(\rho_\pm) + \rho_\pm u_\pm^2 \\ & \Leftrightarrow -s_\varepsilon^2 \rho_\varepsilon + s_\varepsilon^2 \rho_\pm - s_\varepsilon \rho_\pm u_\pm + p_\varepsilon(\rho_\varepsilon) - \mu \frac{s_\varepsilon \rho'_\varepsilon \rho_\pm}{\rho_\varepsilon^2} + \mu \frac{\rho'_\varepsilon \rho_\pm u_\pm}{\rho_\varepsilon^2} \\ & \quad + \frac{1}{\rho_\varepsilon} \left(s_\varepsilon^2 (\rho_\varepsilon - \rho_\pm)^2 + 2s_\varepsilon (\rho_\varepsilon - \rho_\pm) \rho_\pm u_\pm + (\rho_\pm u_\pm)^2 \right) = -s_\varepsilon \rho_\pm u_\pm + p_\varepsilon(\rho_\pm) + \rho_\pm u_\pm^2 \\ & \Leftrightarrow -\mu \frac{\rho'_\varepsilon}{\rho_\varepsilon^2} \rho_\pm (s_\varepsilon - u_\pm) = p_\varepsilon(\rho_\pm) - p_\varepsilon(\rho_\varepsilon) + \rho_\pm u_\pm^2 + s_\varepsilon^2 \rho_\varepsilon - s_\varepsilon^2 \rho_\pm \\ & \quad - \frac{1}{\rho_\varepsilon} \left(s_\varepsilon^2 \rho_\varepsilon^2 - 2s_\varepsilon^2 \rho_\varepsilon \rho_\pm + s_\varepsilon^2 \rho_\pm^2 + (2s_\varepsilon \rho_\varepsilon - 2s_\varepsilon \rho_\pm) (\rho_\pm u_\pm) + \rho_\pm^2 u_\pm^2 \right) \\ & \Leftrightarrow -\mu \frac{\rho'_\varepsilon}{\rho_\varepsilon^2} \rho_\pm (s_\varepsilon - u_\pm) = p_\varepsilon(\rho_\pm) - p_\varepsilon(\rho_\varepsilon) + \rho_\pm u_\pm^2 - s_\varepsilon^2 \rho_\pm + 2s_\varepsilon^2 \rho_\pm - \frac{s_\varepsilon^2 \rho_\pm^2}{\rho_\varepsilon} - 2s_\varepsilon \rho_\pm u_\pm \\ & \quad + 2s_\varepsilon \frac{\rho_\pm^2 u_\pm}{\rho_\varepsilon} - \frac{\rho_\pm^2 u_\pm^2}{\rho_\varepsilon} \\ & \Leftrightarrow -\mu \frac{\rho'_\varepsilon}{\rho_\varepsilon^2} \rho_\pm (s_\varepsilon - u_\pm) = p_\varepsilon(\rho_\pm) - p_\varepsilon(\rho_\varepsilon) + \rho_\pm u_\pm^2 + s_\varepsilon^2 \left(\rho_\pm - \frac{\rho_\pm^2}{\rho_\varepsilon} \right) + s_\varepsilon \left(\frac{2\rho_\pm^2 u_\pm}{\rho_\varepsilon} - 2\rho_\pm u_\pm \right) \\ & \quad - \frac{\rho_\pm^2 u_\pm^2}{\rho_\varepsilon} \\ & \Leftrightarrow -\rho'_\varepsilon (s_\varepsilon - u_\pm) = \frac{\rho_\varepsilon}{\mu} \left[s_\varepsilon^2 (\rho_\varepsilon - \rho_\pm) + s_\varepsilon (2\rho_\pm u_\pm - 2\rho_\varepsilon u_\pm) + \rho_\varepsilon u_\pm^2 - \rho_\pm u_\pm^2 + \frac{\rho_\varepsilon}{\rho_\pm} (p_\varepsilon(\rho_\pm) - p_\varepsilon(\rho_\varepsilon)) \right] \end{aligned}$$

Therefore,

$$\begin{aligned} \rho'_\varepsilon &= \frac{\rho_\varepsilon}{\mu(s_\varepsilon - u_+)} \left((s_\varepsilon - u_+)^2 (\rho_+ - \rho_\varepsilon) + \frac{\rho_\varepsilon}{\rho_+} (p_\varepsilon(\rho_\varepsilon) - p_\varepsilon(\rho_+)) \right) \\ &= \frac{\rho_\varepsilon}{\mu(s_\varepsilon - u_-)} \left((s_\varepsilon - u_-)^2 (\rho_- - \rho_\varepsilon) + \frac{\rho_\varepsilon}{\rho_-} (p_\varepsilon(\rho_\varepsilon) - p_\varepsilon(\rho_-)) \right). \end{aligned}$$

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